16 (a) Rationalise the denominator of $\frac{22}{\sqrt{11}}$

Give your answer in its simplest form.

$22/\sqrt{11} \times \sqrt{11}/\sqrt{11} = 22\sqrt{11}/11 = 2\sqrt{11}.$

2 marks total
Multiply by √11/√11
Simplify 2√11

(2)

(b) Show that $\frac{\sqrt{3}}{2\sqrt{3}-1}$ can be written in the form $\frac{a+\sqrt{3}}{b}$ where a and b are integers.

$\sqrt{3}/(2\sqrt{3} - 1) \times (2\sqrt{3} + 1)/(2\sqrt{3} + 1)$

$= (6 + \sqrt{3})/(12 - 1) = (6 + \sqrt{3})/11.$

Therefore $a = 6$ and $b = 11$.

3 marks total
Multiply by conjugate
Use √3 × 2√3 = 6
a = 6, b = 11

(3)

(Total for Question 16 is 5 marks)